

LET THE NEURONS DIE

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INTRODUCTION

As machine learning becomes more prevalent for critical applications, the underlying model architecture (e.g., activation function) poses vulnerability to a variety of attacks. **ReLU** is a popular activation function for its simplicity, low computational cost, and performance. However, ReLU can suffer from **dead neurons** which adversaries can abuse.

PROJECT GOALS

With this research project, we present novel attack methods to **exploit ReLU's dead neuron weakness**. Our objective is to cause weights to become negatively biased. By creating zero activations during the forward step, there is a resultant zero gradient update.

We propose:

- A novel dynamic **Data Ordering Attack (DOA)** using weight decay rather than loss (inspired by [1])
- Multi-Layer **Data Poisoning Attacks** using crafted data from the repurposed **Inverting Gradients (IG)** approach [2]
 - Using weights from **Soft Knockout Attack (IG-SKA)** [3]
 - Using weights from dynamic **Data Ordering Attack (IG-DOA)**

THREAT MODELS

Point of Attack: Prior to training

Attack Preparation Phase:

White-box access to victim model architecture, dataset and initial weights

Attack Execution Phase:

- **DOA:** requires NO new data injection
- **IG-SKA & IG-DOA:** requires new poisoned data injection

METHODOLOGY

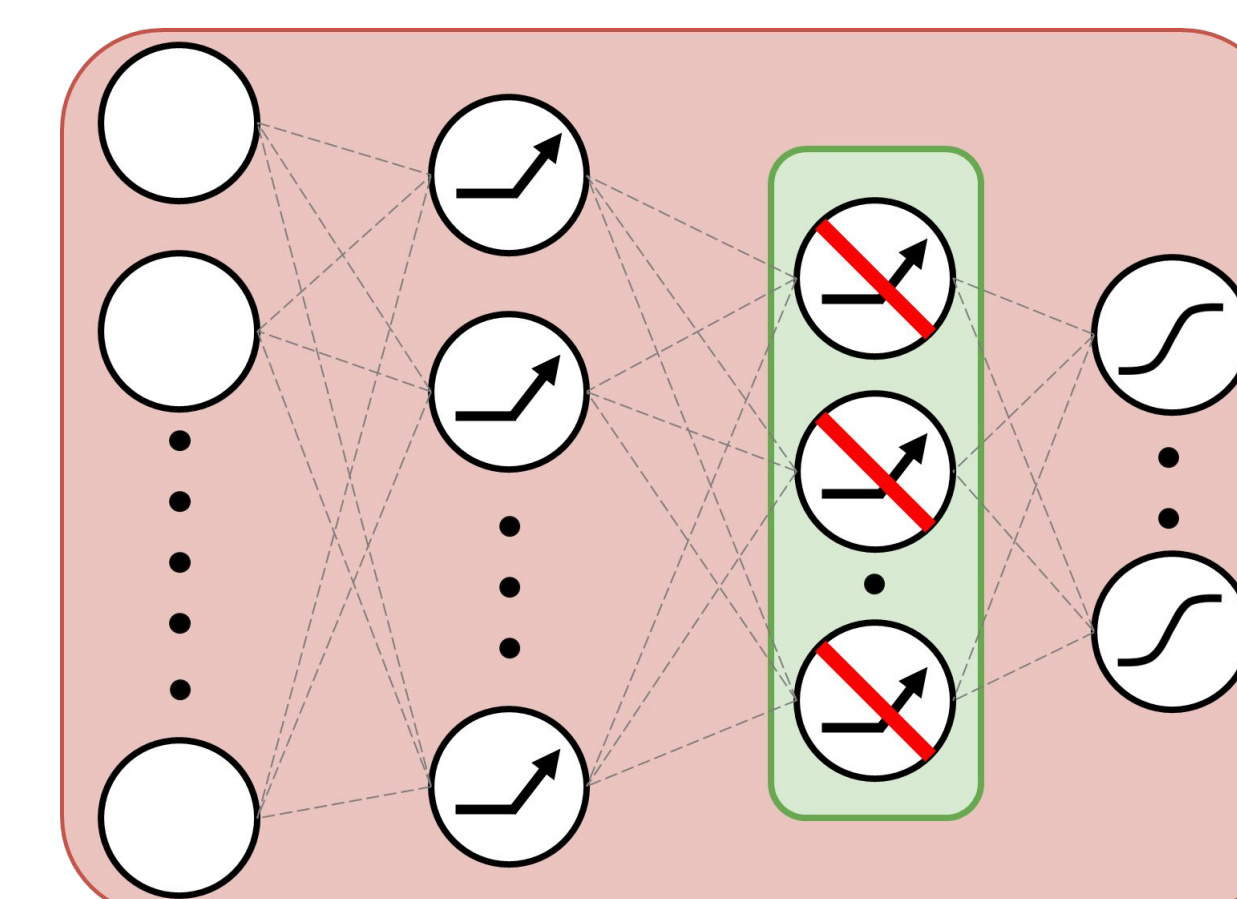
Dynamic Data Ordering Attack (DOA)

Input: Victim model's initial weights, $\mathbf{W}_i$

Idea: A target layer ℓ with ReLU activations is targeted to minimize weights

1. Train data points (1...n) individually from $\mathbf{W}_i$
2. Calculate $\text{score}_k = \sum \mathbf{W}_{k\ell}$ = Sum of weights for layer ℓ after training on point $\mathbf{k}$
3. Add data point $\mathbf{k}$ with lowest score to candidate list
4. Repeat 1-3 starting from $\mathbf{W}_k$

Output: Candidate list of data ordered points negatively biasing layer ℓ



Multi-Layer Data Poisoning Attacks

Frontend Attack

Option 1: SKA [3]

Input: Victim model's initial weights, $\mathbf{W}_i$

Idea: e.g., for every 2 consecutive layers with ReLU, weights $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{B} \in \mathbb{R}^{\ell \times m}$ and biases $\mathbf{a} \in \mathbb{R}^m$, $\mathbf{b} \in \mathbb{R}^\ell$, $y = \text{ReLU}(\mathbf{B} \text{ReLU}(\mathbf{A}\mathbf{x} + \mathbf{a}) + \mathbf{b})$, we concentrate +ve components and cross $\mathbf{A}$ & $\mathbf{B}$, aiming to annihilate $\mathbf{W}$

$$\text{ReLU}\left(\begin{matrix} \text{matrix A} \\ \vdots \\ \text{matrix B} \end{matrix} \begin{matrix} \vdots \\ x \end{matrix} + \begin{matrix} \vdots \\ a \end{matrix}\right) + \begin{matrix} \vdots \\ b \end{matrix} = \begin{matrix} \vdots \\ y \end{matrix}$$

Output: Harmful weights impeding training, $\mathbf{W}_t$

Option 2: DOA

Output: Weights from model trained with ordered dataset, $\mathbf{W}_t$

Frontend Attack
Option 1: SKA
Option 2: DOA

Model Weights

Backend Attack
IG

Backend Attack

IG

Original IG Setting [2]:

- In **Federated Learning**, each user trains a model on a server with **private** local data. They receive current weights from the server and share parameter updates (gradients)
- Original Goal: Training data reconstruction from gradient information (privacy infringement)

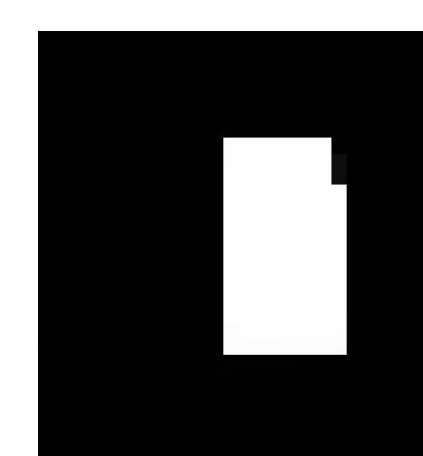
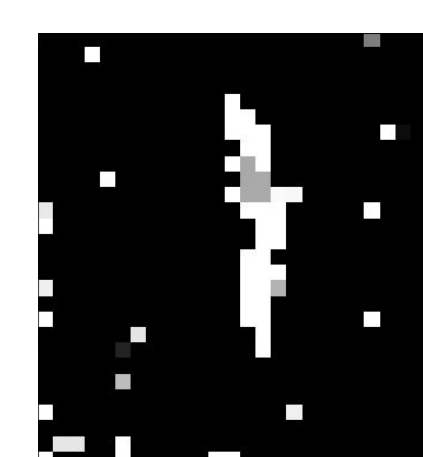
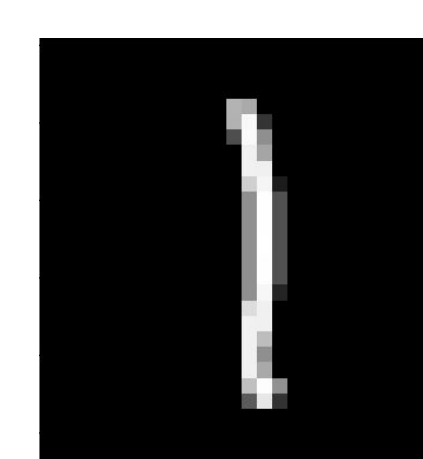
IG-SKA & IG-DOA

Input: Victim model & $\mathbf{W}_t$

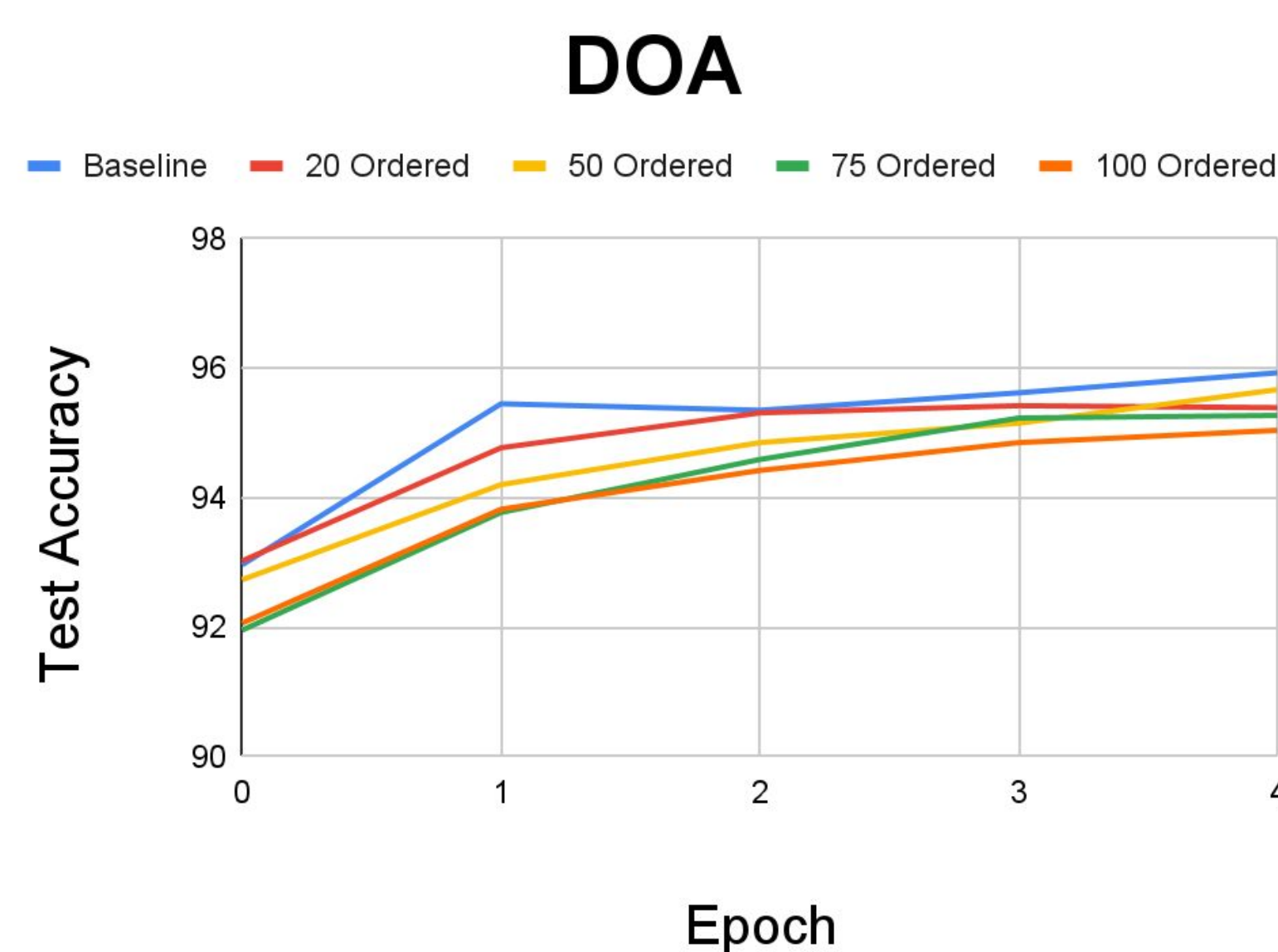
Idea: Initialize FL model with $\mathbf{W}_t$ and **repurpose** IG to construct poisoned input data

Output: Poisoned data sample(s) for every class

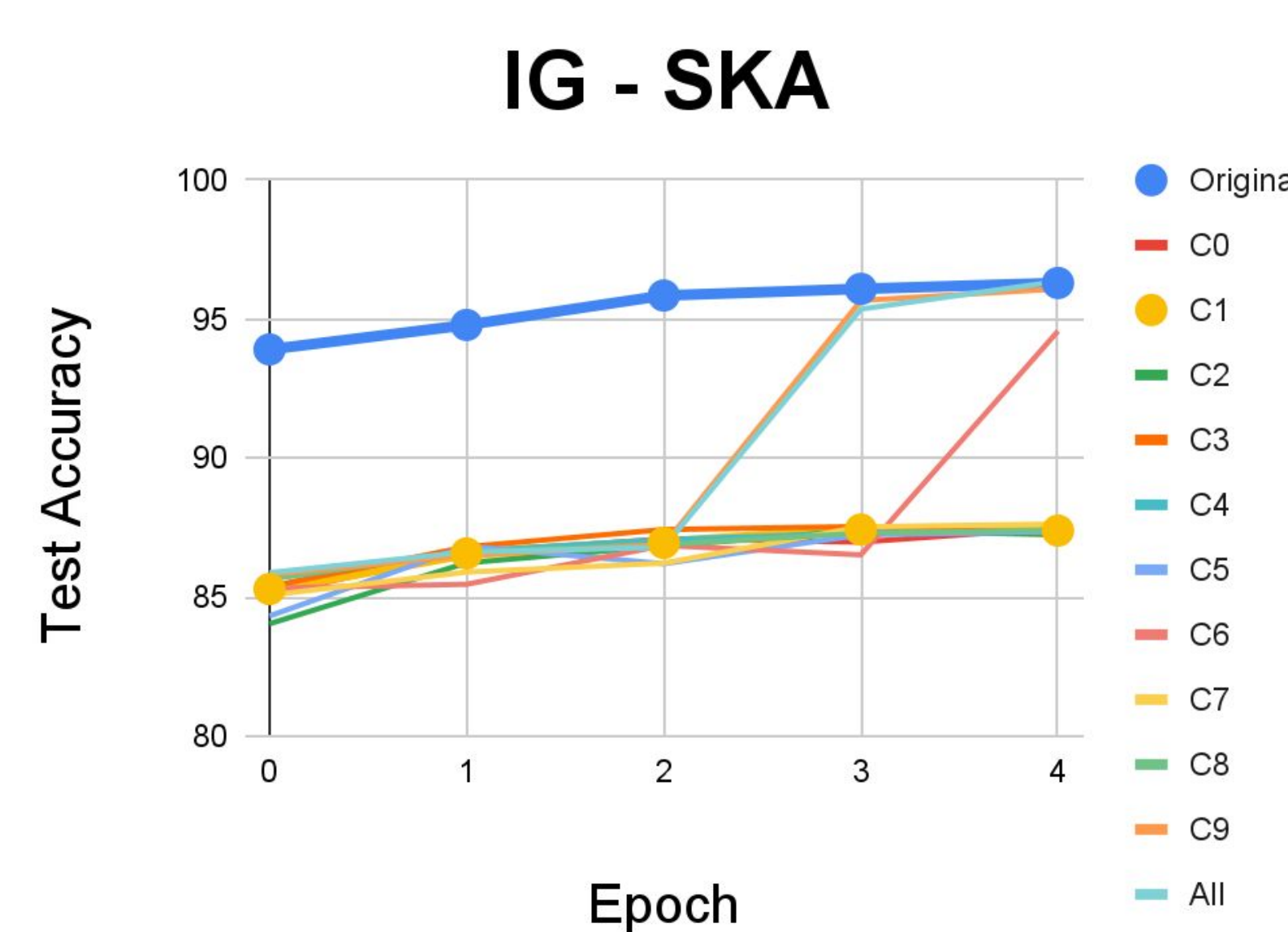
Poisoned Samples



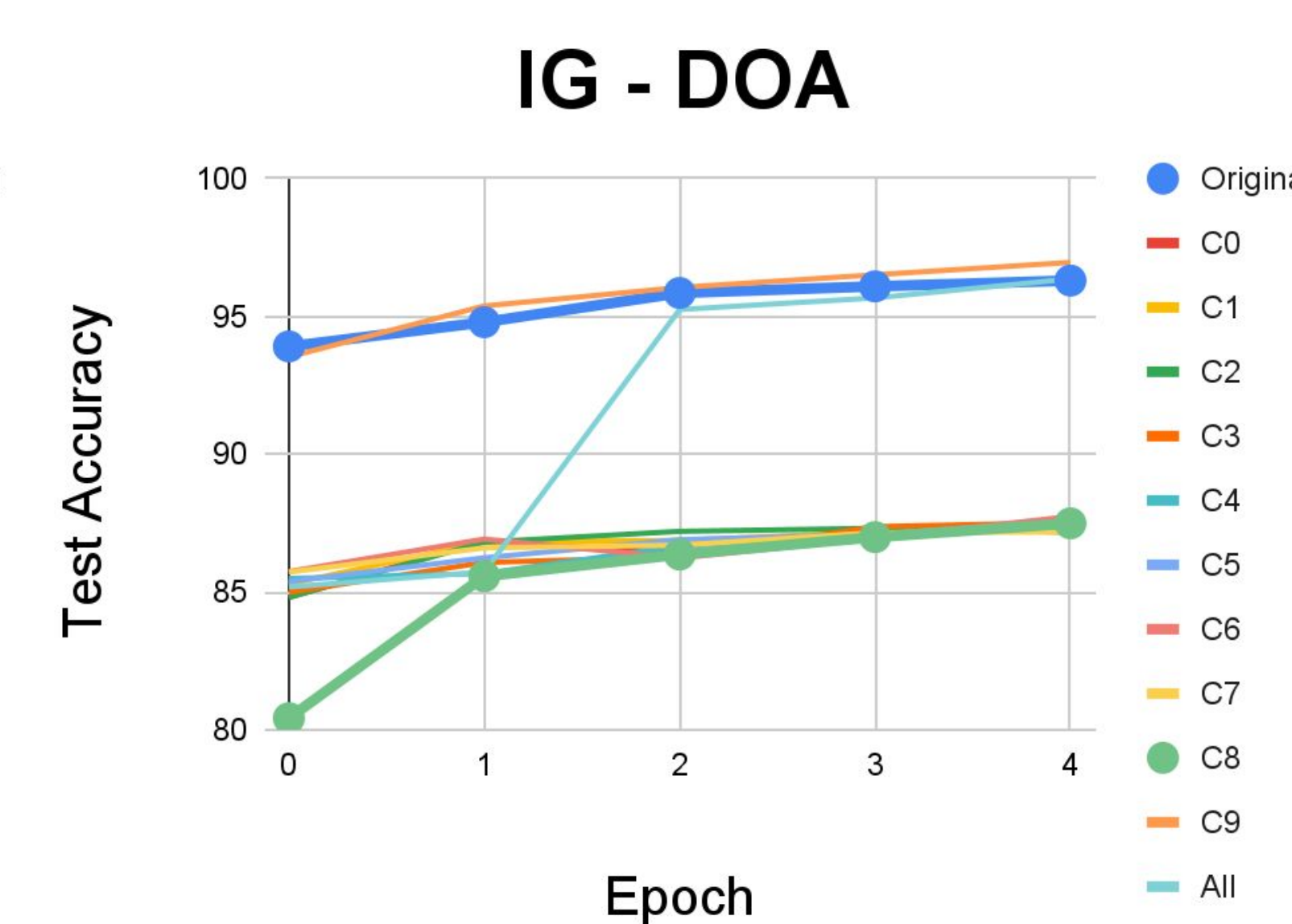
RESULTS



Ordered indicates the accuracy on the test dataset at each epoch when ordering # of data points during training



C# indicates the accuracy on the test dataset at each epoch when adding 200 poisoned samples reconstructed from Class # into the training dataset (All indicates poisoning all classes)



DISCUSSION

Dynamic Data Ordering Attack

- + Difficult to detect
- + Minimally invasive as only a small subset of data points need ordering
- Smaller performance degradation
- Computationally expensive

Multi-Layer Data Poisoning Attacks

- + Aggressive performance degradation
- + Can craft many adversaries
- More obvious to detect

IG-SKA

- + Has higher potential in the best case
- Mathematically derived, and deviating from exact values limits potential gain

IG-DOA

- + More consistent in average cases
- Computationally expensive
- Easier to detect

FUTURE WORK

Attack Effectiveness:

- Study the correlation of #samples ordered/poisoned vs accuracy drop
- Maintain or exceed attack quality with fewer ordered/poisoned samples

Attack Extensibility & Scalability:

- Test on more model architectures (e.g., CNNs, ResNet, etc.)
- Test with more datasets (e.g., CIFAR, ImageNet, etc.)
- Translate to other domains (e.g., NLP)

REFERENCES

- [1] I. Shumailov et al., "Manipulating sgd with data ordering attacks," Advances in Neural Information Processing Systems, vol. 34, pp. 18021–18032, 2021.
- [2] J. Geiping, H. Bauermeister, H. Dröge, and M. Moeller, "Inverting gradients-how easy is it to break privacy in federated learning?," Advances in Neural Information Processing Systems, vol. 33, pp. 16937–16947, 2020.
- [3] Grosse, K., Trost, T. A., Mosbach, M., Backes, M., and Klakow, D. On the security relevance of weights in deep learning. arXiv preprint arXiv:1902.03020, 2020.